

Rakshita Gupta

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EDUCATION

Purdue University | B.S. Computer Engineering Technology

Expected May 2028

Minors: Psychology and ECE + Entrepreneurship Certificate

GPA: 3.40

Relevant Coursework: Digital System Design, Data Structures, Advanced C Programming, Electrical Engineering Fundamentals I, Python for Data Science, Systems Development

EXPERIENCE

DirtView Inc. | *XR / AR Product Intern*

Remote · Sep 2026 – Present

- Developing a platform that brings XR smart glasses to construction management (DirtVision)
- Researching XR/AR wearable platforms and SDKs; writing requirements, user stories, and test plans
- Writing test scenarios and documentation; supporting prototype and interaction design

Digital Realty | *Software Engineering Extern*

Remote · Jun – Jul 2026

- Developed a Python and Word VBA (Visual Basic for Applications) tool to automate the extraction and conversion of complex tables from HIPAA-compliance report Word documents into Excel spreadsheets.
- Designed an intuitive interface enabling legal professionals to efficiently review structured data and seamlessly append company recommendations, saving an estimated **2–3 months** of manual data entry.
- Operated under a strict NDA with a locked-down, access-restricted document that could not be shared with any external or cloud services, building and testing the entire automation locally within the client's secure environment.

Motio | *UI/UX Design Extern*

Remote · May – Jul 2026

- Redesigned an internal data-retention console from the ground up: 6 screens plus the app header as a clickable prototype with a React handoff spec
- Built a 55-token light/dark design system with a Motio-blue palette and 3 typefaces
- Made destructive actions safer: type-the-record-count confirmation with paste blocking, a live match hint, and an animated READY → PURGING state change
- Added power-user features (command palette, keyboard shortcuts with a help overlay, Undo toasts, sortable and filterable tables, density toggle)
- Kept an accessibility baseline: visible focus rings, reduced-motion support, ARIA, skip link, full keyboard operation

BlueBubbles (Open Source) | *UI/UX Design Contributor*

Remote · Jun 2020 – Aug 2023

- Contributed UI/UX design to BlueBubbles, an open-source Flutter/Dart application bringing iMessage to Android phones and Windows/Linux computers, with **100K+** downloads and a 4.6-star rating on Google Play.
- Streamlined the full chat interface and co-designed the app's robust theming engine — now a headline feature of the Play Store listing.
- Worked on an app complete with tapbacks, GamePigeon support, and FaceTime calls.
- Worked directly with the 5-person core developer team, iterating on feedback from a 13,000+ member Discord community and GitHub beta testers.

CPR Cell Phone Repair | *Electronics Technician & Sales Associate*

Augusta, GA · Sep 2022 – Aug 2023

- Completed ~**200** repairs per month across smartphones, tablets, laptops, and game consoles — screen and battery replacements, software repairs, and board-level soldering.
- Handled the ticket system and upsold screen protectors and phone cases to **80%** of customers.
- Lead responsibility for software-based repairs; ranked top of the shop in service speed and customer satisfaction, earning the highest tips among technicians.

CowShopCo (Etsy) | *Founder & Maker*

Augusta, GA / West Lafayette, IN · Apr 2020 – Present

- Design, 3D-print, and sell an original product catalog — jewelry, charms, and household items — modeled entirely in Blender and Fusion 360; **100+** orders shipped with a 5-star Etsy rating.
- Run the business end to end (product photography, listings, pricing, shipping, and customer service) and place products in local businesses and fairs on commission.

GERI Summer Residential Camp, Purdue University | *Summer Camp Counselor*

West Lafayette, IN · Jul 2025

- Directly mentored 12 gifted students (primarily 9th–12th grade girls) and managed an entire residence-hall floor at a two-week academic summer camp of 300+ campers.
- Taught daily personal and life-skills classes and served as a live-in first point of contact for conflict resolution across the camp.

CDI Official | *Founder*

Augusta, GA · Nov 2014 – Jun 2023

- Founded a media production studio delivering 50+ professionally edited videos plus apps, websites, event videography, and photography that built clients' brand identities and online presence, including healthcare clients under NDA.
- Produced immersive 360°/VR simulation videos for the Augusta University Medical Simulation Center, later adapted by the department into interactive VR training apps for medical students.
- Filmed and edited clinical procedure explainers for AugustaENT patients (e.g., Ultrasound-Guided Fine-Needle Aspiration Biopsy, pediatric surgery prep), credited with measurably easing patient anxiety.

LEADERSHIP

Embedded Systems @ Purdue (ES@P) | *Project Manager, IntelliThings* West Lafayette, IN · Aug 2026 – Present

- Co-leading a 25-member, 3-subteam team building an LLM-powered smart home system that needs no hand-written rules
- Running a 13-week build plan and weekly 3-hour work sessions toward a demo at the ECE SPARK Challenge; screened 72 applicants to staff the team
- Stack: Embedded C, ESP-IDF, FreeRTOS, MQTT, ESP32 sensor nodes, Home Assistant with MCP, Matter, GPT-4.1 API

PROJECTS

CowBubbles | *Flutter · Dart · Kotlin · GLSL* | github.com/rakkicow/CowBubbles 2026 – Present

- Redesigned the UI of OpenBubbles, the iMessage-for-Android fork of BlueBubbles (UI/UX contributor, 2020–23): 50 screens restyled; v0.1-alpha APK released
- Built a 13-file design system: OKLCH-to-sRGB color math, FNV-1a-hashed per-chat hues, liquid-glass widgets
- Streamed album-art pixels from a Kotlin listener to Dart to drive a GLSL shader background and synced lyrics

CarLyrics | *Kotlin · Android Auto · Car App Library · AGSL* Sep 2026 – Present

- Built an Android Auto app showing word-synced lyrics for any music player via MediaSession; working prototype
- Wrote a per-frame position-extrapolating sync engine and TTML/JSON/YRC/LRC parsers with JUnit tests
- Rendered a normal Android View onto the car surface via NavigationTemplate, with an AGSL album-art shader

TaskReceipt | *Python · Bluetooth LE (bleak) · Flask · Google Tasks API* Aug 2026

- Reverse-engineered a ~\$6 Bluetooth receipt printer's BLE protocol; fixed 3 errors in community docs
- Prints Google Tasks to-dos and weather from a Flask control panel; runs as a code-signed macOS background job

ESP32 Now-Playing Clock | *C++ (Arduino-ESP32) · FreeRTOS · PlatformIO · TFT_eSPI · FastLED* Sep 2026 – Present

- Built a LilyGO T-Dongle-S3 (ESP32-S3) dongle showing now-playing Last.fm tracks with album art on a 160×80 LCD
- Pinned network I/O to a FreeRTOS task on core 0 so rendering on core 1 never stalls
- Color-matched gradient and LED glow per track, standby clock when idle; documented 8 hardware/library findings

Lunar Lander on FPGA (ECE 27000) | *SystemVerilog · Lattice iCE40HX8K · Verilator · Yosys* Spring 2026

- Designed a lunar-lander game in SystemVerilog on a Lattice iCE40HX8K FPGA: memory, arithmetic, control, display
- Built all math in BCD with no +/− operators, from a full adder up to a 4-digit ten's-complement adder/subtractor
- Verified with Verilator C++ testbenches and Icarus Verilog simulation; synthesized with Yosys
- Built an 11-state Moore FSM sequence lock (8-bit shift-register passcode); demoed on the FPGA

River-Crossing Path Solver (ECE 36800) | *C99 · 0-1 BFS · valgrind* Spring 2026

- Built a 0-1 BFS solver in C over an implicit graph of ~2MN board positions, using a circular-buffer deque and parent-array path reconstruction
- Seeded the search from every right-bank position at once, returning each starting row's minimum turns in a single pass
- C99 compiled with -Wall -pedantic -O3, graded under valgrind and time limits

Mnemoo (in design) | *Android XR · ML Kit · FaceNet · Gemini Nano (planned)* Jul 2026 – Present

- Designing Android XR glasses for dementia care with a HUD that names the face of the person you see, details your last interaction, and recaps the conversation. Like a second brain
- Architected an all-on-device stack: ML Kit face detection, FaceNet-style embeddings, on-device speech, Gemini Nano summaries
- Privacy by design: camera and mic wake on request only, stays quiet when unsure, no cloud; phone app first, then XR headset, then glasses

3-Band Audio Equalizer (ECE 20007) | *LTspice · op-amps · LM386 · Keysight oscilloscope* Spring 2026

- Designed a 3-band analog EQ (filters, op-amp stages, summing amp, LM386) delivering 894 mW into 8 Ω
- Verified the band cutoffs and output ripple against spec on a Keysight oscilloscope

FormatScope, Purdue Chips & AI Hackathon (in progress) | *PyTorch · Verilog · cocotb · Yosys · OpenSTA · sky130* Sep 2026

- Compare ResNet-8 accuracy against sky130 chip area and delay across 5 number formats on a 3-person team
- Own synthesis, OpenSTA timing, the formatscope CLI, CI, and GPU runs

Personal Portfolio Website | *Astro · Tailwind CSS · GSAP · Lenis · GitHub Pages* | rakshita.me Oct 2023 – Present

- Designed and built rakshita.me with Astro, Tailwind CSS, GSAP, and Lenis with scroll-driven animations, light/dark theming, and interactive easter eggs in a modern, fully responsive design.
- Deployed via GitHub Pages with custom domain routing and DNS configuration.

TECHNICAL SKILLS

Languages: C, C++, Python, Kotlin, Dart, Swift, SystemVerilog/Verilog, JavaScript/TypeScript, HTML/CSS, Java, SQL, VBA, MATLAB

Hardware & Embedded: FPGA (iCE40), ESP32, FreeRTOS, PlatformIO/Arduino, Bluetooth LE, KiCad, LTspice, Verilator, Yosys, oscilloscope, soldering

Software & Tools: Flutter, Android Auto, GLSL/AGSL shaders, Astro, React, Tailwind, Flask, JUnit, Git/GitHub Actions, Linux, Claude Code

Design: Figma, design systems, prototyping, accessibility, Blender, Fusion 360, 3D printing, Final Cut Pro, Logic Pro, full Adobe Suite